

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | *Bengaluru, India* Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients**, processing **10,000+ monthly submissions** through REST integrations with Mayden's iaptus platform; contributed to **\$340K+** in revenue.
- Architected a full-stack data annotation platform (React, Node.js, MongoDB) with clean data models and role-based APIs, cutting manual work by **100+ hours/month** and lifting labeling throughput by **60%**.
- Investigated production incidents during releases and client integrations to root cause and shipped fixes; implemented AWS KMS encryption, auth hardening, rate limiting, and integration tests, remediating **20+ VAPT findings**.
- Served as primary technical POC for the eTriage backend; onboarded and mentored a second developer to on-call readiness and piloted a COO-sponsored GitHub Copilot AI-assisted-development experiment.

Cario Growth Services, Backend Developer Intern | *Chennai, India* Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, **PostgreSQL**) with input validation, clean data models, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM auto-commenting feature (**4-bit quantized Llama-7B via llama.cpp**) with async queued generation and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | *Remote* Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN platform with a multi-branch data model, Stripe hybrid payments, and cron automation; live **3+ years**.
- Delivered a second MERN client engagement (US-based inventory management system) end-to-end as a paid freelance build.

Projects

Spark ETL Performance Optimization (GitHub) PySpark, Spark UI, SQL, NYC TLC Dataset

- Built and tuned a Spark ETL pipeline on a **~1.4B-row** dataset, extracting, transforming, and loading trip data with broadcast joins, adaptive execution, skew-join handling, schema-aware reads, partitioning, and column pruning to cut end-to-end runtime **~25%** (17–19 min to 14.3 min) and the analytics tail-latency ratio from **2.14x to 1.74x**, profiled via Spark UI.

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a Python/FastAPI backend service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, deterministic extraction with LLM fallback, and output verification, returning cell-level provenance with SQLite-backed storage; hardened with **150+ automated tests** and provider routing across OpenAI/Groq.

Sonare – Offline Sign / Speech App (GitHub) React, Electron, FastAPI, MediaPipe, whisper.cpp

- Engineered a FastAPI backend and Electron/React frontend integrating camera, microphone, and local services (MediaPipe hand tracking, whisper.cpp ASR, local TTS) into one offline system, designing async APIs and queues to keep the UI responsive under local video and audio streams.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: JavaScript, TypeScript, Python, SQL, C++, Java, PostgreSQL, MongoDB, SQLite

Backend & APIs: Node.js, Express, Fastify, FastAPI, REST APIs, integration testing, system integrations

Data & ETL: PySpark, Spark, ETL pipelines, data modeling, partitioning, Hadoop / MapReduce

Tools & Practices: Docker, Git, GitHub Actions, CI/CD, pytest, Agile / Scrum, AI-assisted dev (Copilot)

Cloud / Platforms: AWS, GCP, Azure, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.